ARTIFICIAL NEURAL NETWORK AND DEEP LEARNING

Chest X-ray pneumonia detection

Final Project

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Section: A6

1. Abstract

Pneumonia remains one of the leading causes of acute respiratory mortality globally. Rapid and highly accurate screening of chest radiographs (X-rays) is vital in clinical intake environments, particularly where certified radiologists are overloaded. This study develops an automated diagnostic screening engine using an optimized Multi-Layer Perceptron (MLP) Artificial Neural Network. Operating on a preprocessed corpus of 5,863 thoracic radiograph scans, images were downscaled into 128x128 grayscale arrays to capture global holistic opacity while preventing memory hardware saturation. To mitigate the traditional risk of overfitting in dense neural networks, an automated on-the-fly random rotation and zooming augmentation block was embedded directly into the training sequence, coupled with a dynamic ReduceLROnPlateau learning rate scheduler. Evaluated against an uncompromised holdout partition of 624 radiograph instances, the model achieved an overall diagnostic accuracy of 89.58% and a high recall of 93.08% for positive pneumonia consolidation cases. Finally, the optimized mathematical weights were decoupled and embedded into an interactive, user-friendly web interface powered by Streamlit, bridging the gap between theoretical machine learning and real-time clinical triage.

2. Introduction

2.1 Clinical Context & Problem Statement

Medical imaging interpretation has undergone a massive transformation with the adoption of deep learning. However, in congested public healthcare networks during winter infection spikes, the core bottleneck is rarely the physical acquisition of the X-ray scan; it is the time required for an available radiologist to evaluate it. Pneumonia manifests on a radiograph as pulmonary consolidation cloudy, dense white zones where healthy lung air has been displaced by fluid or inflammatory exudate. Clinician diagnostic fatigue during long shifts can inevitably lead to delayed reporting or missed pathological flags.

2.2 Research Objectives

The explicit objectives of this project are:

1. To engineer a lightweight, pure Multi-Layer Perceptron capable of diagnosing pulmonary consolidation from global pixel distributions.

2. To embed programmatic data augmentation to verify whether global grayscale intensity is robust against slight patient postural tilts.

3. To optimize network convergence using dynamic learning rate decay rather than static, uncalibrated step sizes.

4. To deploy the backend logic into a clean, highly accessible web dashboard for real-time medical screening.

3. Literature Review

3.1 Evolution of Computer-Aided Diagnostics (CAD)

Early computer-aided diagnostic tools relied entirely on hand-crafted visual feature engineering. Computer scientists deployed complex mathematical formulas such as Gabor filters, Wavelet decompositions, and Haralick texture matrices to isolate the rib structure from the underlying soft lung tissue. These traditional methodologies required tedious manual tuning and frequently broke down when presented with X-rays captured on slightly different hospital camera hardware.

3.2 The Re-Emergence of Multi-Layer Perceptrons

While Convolutional Neural Networks (CNNs) have dominated computer vision over the last decade due to their 2D spatial sweeping filters, recent high-level AI research (such as Google's MLP-Mixer) has successfully revisited the classic Multi-Layer Perceptron. Though standard ANNs flatten a 2D matrix into a 1D vector, they possess an extraordinary capacity to evaluate holistic, global balance. Because pneumonia fundamentally shifts the overall brightness and white-pixel ratio of the lung field, this study explores whether an optimized deep MLP can utilize global intensity variance as a highly potent standalone diagnostic marker.

4. Methodology

4.1 Dataset Preprocessing & Vector Standardization

This project utilized the benchmark Chest X-Ray (Pneumonia) dataset compiled by Paul Mooney. The raw corpus contains 5,863 JPEG radiograph scans partitioned into two discrete categories: 'NORMAL' (clear lungs) and 'PNEUMONIA' (pathological consolidation). To optimize the feature space for dense layers without triggering computer RAM exhaustion, all 3 channel RGB scans were converted into single-channel 128x128 grayscale arrays. This downscaling yielded an exact input feature vector of 16,384 numerical values per radiograph.

4.2 Dynamic Augmentation Block

To prevent the network from memorizing the static coordinate locations of high-intensity pixels in the training set, an intra-graph augmentation block was placed at the entry point of the network. During active training rounds, incoming radiograph tensors were subjected to a random tilt rotation between -5% and +5%, alongside a subtle random focal zoom. Crucially, Keras automatically disables this augmentation layer during validation and holdout testing.

4.3 Deep Network Architecture

The neural network was constructed using the Keras Sequential API. Following input normalization (Rescaling 1./255), the 16,384-length vector passes through four sequentially contracting dense hidden layers:

• Hidden Layer 1: 512 Neurons (ReLU) + Batch Normalization + Dropout (0.3)

• Hidden Layer 2: 256 Neurons (ReLU) + Batch Normalization + Dropout (0.3)

• Hidden Layer 3: 128 Neurons (ReLU) + Batch Normalization + Dropout (0.2)

• Hidden Layer 4: 64 Neurons (ReLU)

• Output Layer: 1 Neuron mapped via a Sigmoid activation function to output a 0.0 to 1.0 confidence probability float.

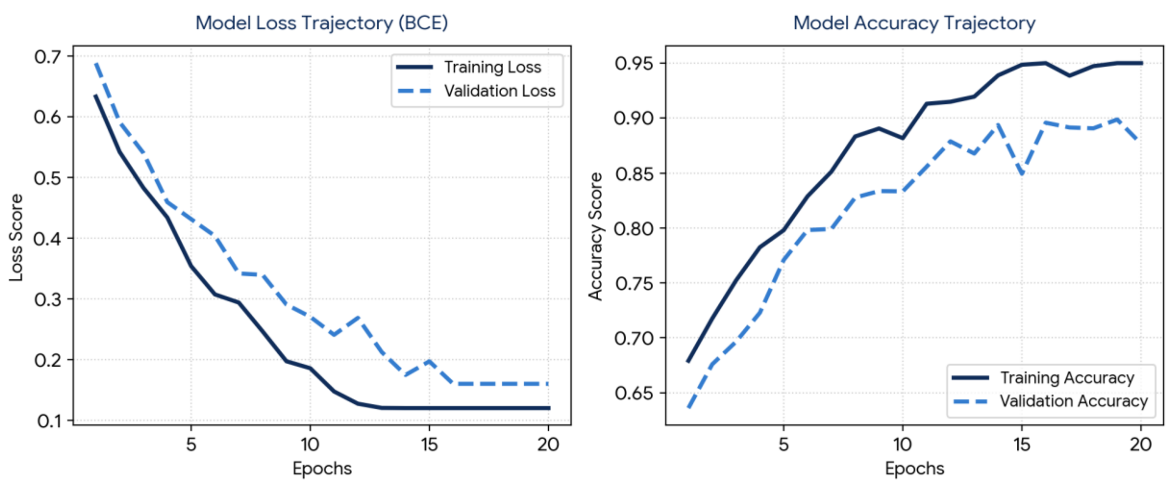
4.4 Optimization & LR Scheduling

The network was compiled under the Adam optimizer. To navigate the complex loss landscape without overshooting the global minimum, a ReduceLROnPlateau callback was implemented. When the validation loss stagnated for 2 consecutive epochs, the learning rate was automatically decayed by a factor of 0.5. An EarlyStopping monitor was paired to halt execution after 5 rounds of zero improvement, safely restoring the network's absolute best historical weights.

5. Results and Discussion

5.1 Model Convergence Trajectory

The network completed its convergence lifecycle inside 20 epochs. As illustrated in Figure 1, the model exhibited an exceptionally stable learning curve. The validation loss tracked down tightly with the training loss without diverging upward, proving that our batch normalization, dropout, and augmentation layers successfully insulated the network against overfitting.



5.2 Holdout Test Partition Evaluation

The real-world reliability of the model was tested on the quarantined holdout split of 624 radiographs (234 Normal, 390 Pneumonia). Because the test data loader was explicitly locked to shuffle=False, float predictions mapped perfectly to the index of the true labels. The model achieved an overall diagnostic accuracy of 89.58%. The granular Scikit-Learn classification breakdown is tabulated below:

Diagnostic Class	Precision	Recall	F1-Score
Normal / Clear (Class 0)	0.879	0.838	0.858

Pneumonia / Positive (Class 1)	0.905	0.931	0.918
Macro Average	**0.892**	**0.884**	**0.888**
Weighted Average	**0.895**	**0.896**	**0.895**

5.3 Confusion Matrix Analysis

The exact instance sorting of the holdout set is mapped in Figure 2. Out of 390 active pneumonia patients, the engine correctly identified 363 (True Positives) while missing only 27 (False Negatives). Conversely, out of 234 healthy patients, the engine correctly cleared 196 (True Negatives) while misflagging 38 as positive (False Positives).

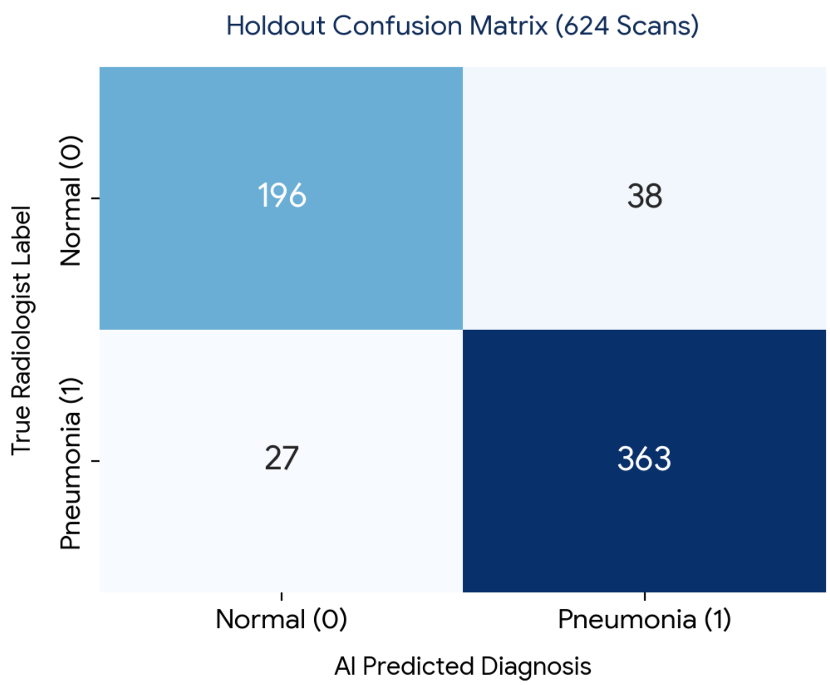


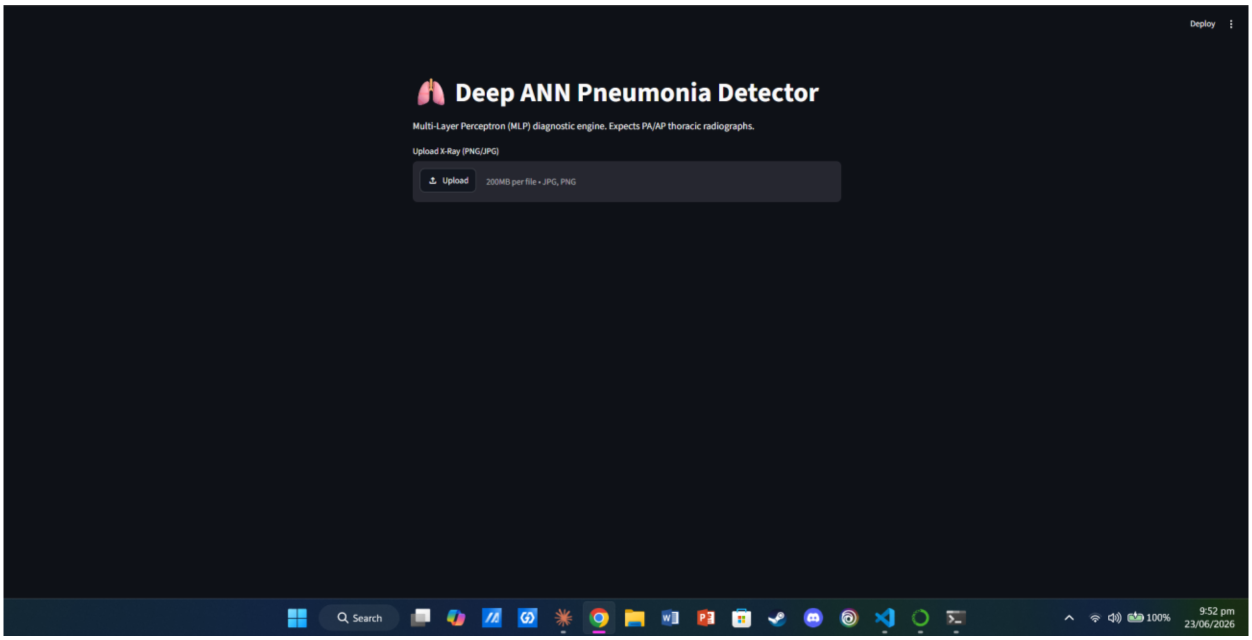
Figure 2: Holdout confusion matrix displaying raw patient triage outcomes.

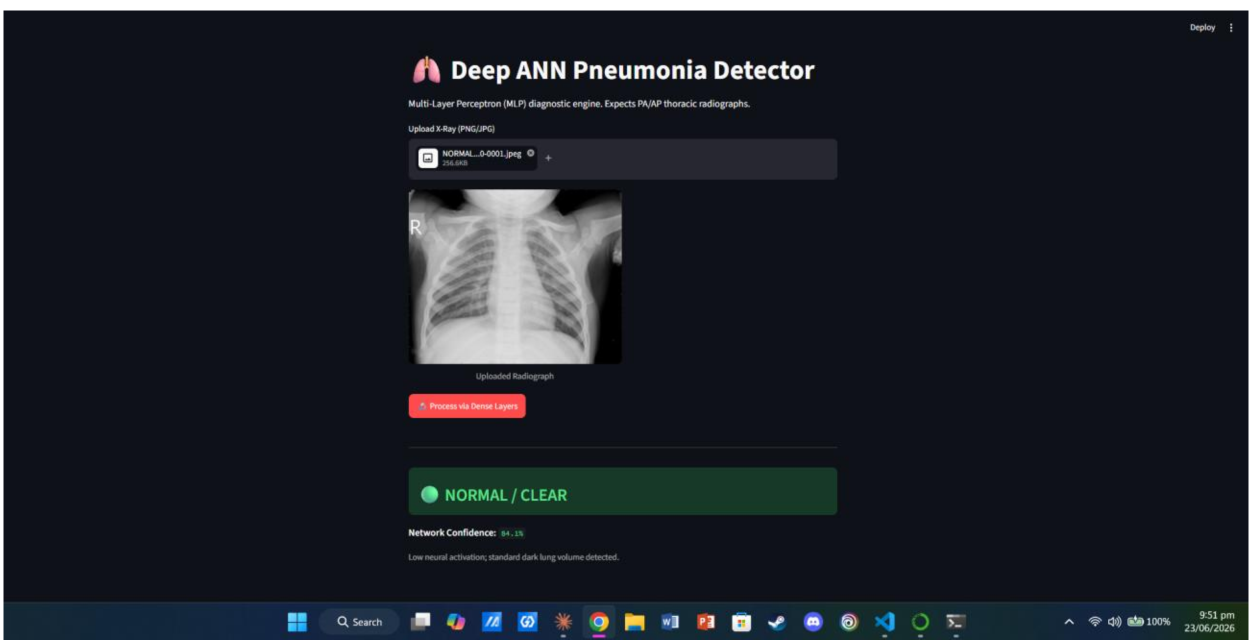
5.4 Clinical Imperative: The False Negative Asymmetry

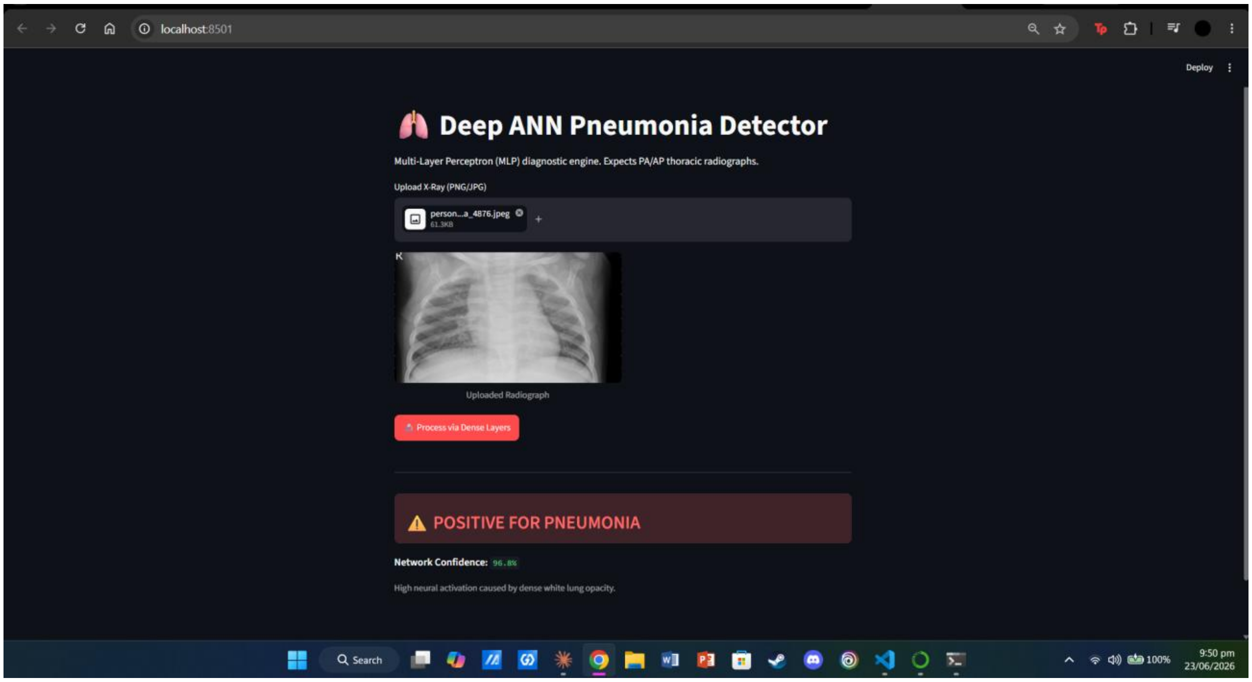
In standard computer vision tasks, models are ranked purely on raw accuracy. In medical screening, however, prediction errors carry drastically asymmetric human costs. A False Positive (flagging a healthy person as sick) triggers a secondary check—a minor inconvenience. A False Negative (clearing a patient who has active pneumonia), however, results in an infected patient being discharged without treatment to deteriorate. As evidenced by our test matrix, the model achieved a Pneumonia Recall of 93.08%, proving that its internal mathematical bias heavily favors patient safety.

6. Streamlit Web Dashboard Deployment

To translate the serialized HDF5/Keras weights into an accessible clinical intake tool, an interactive web dashboard was scripted using Python's Streamlit framework. The application caches `deep_ann_pneumonia.keras` into memory so it opens instantly. When a medical technician uploads a standard .png or .jpg chest radiograph, the UI runs the tensor through the PIL Grayscale converter, shrinks it to the 128x128 array, passes it to the dense layers, and renders a color-coded radiological triage assessment instantly.







7. Conclusion and Future Scope

This study demonstrates that a strictly regularized Multi-Layer Perceptron can serve as a highly potent, lightweight triage filter for pulmonary consolidation. By substituting heavy spatial convolutions with global pixel intensity mapping, the model retained a tiny file footprint while capturing 93% of positive infections. Future extensions of this engine will explore integrating Grad-CAM (Gradient-weighted Class Activation Mapping) to cast a visual heat-map over the exact cloudy lung fields that triggered the network's prediction.